Phase 19: Change Request Log & Change Governance

Document Code: CR-LOG-001

Project: Duplicate Bin Ecosystem

Methodology: Formal Change Control Board (CCB) Process

Notice: Scenarios below include realistic simulated examples clearly marked as **[SIMULATED INTERVIEW SCENARIO]** for interview demonstration.

1. Formal Change Control Procedure

When a requirement change is proposed during the software development lifecycle, the Business Analyst drives the structured change management governance framework:

MERMAID DIAGRAM

```
flowchart TD
    CR_Submit[1. Change Request Submission via CR Form] --> BA_Assess[2. BA Impact & Feasibility Assessmer
    BA_Assess --> Tech_Eval[3. Technical & Architectural Review with Tech Lead]
    Tech_Eval --> Cost_Eval[4. Cost, Schedule & Effort Estimation]
    Cost_Eval --> CCB_Review{5. Change Control Board Review}
    CCB_Review -->|Rejected| Inform_Reject[6a. Document Rationale & Notify Requester]
    CCB_Review -->|Deferred| Backlog_Defer[6b. Log to Future Product Roadmap]
    CCB_Review -->|Approved| Update_Baseline[6c. Update BRD, FRD, RTM & Sprint Backlog]
    Update_Baseline --> Implement[7. Engineering Implementation & UAT Re-verification]
```

2. Change Request Ledger

CR ID	Change Title	Requester	Classification	Impact Summary	Decision Status	Decision Date
CR-001	Transition from SQLite to PostgreSQL for SaaS Backend	Lead Architect	Technical Infrastructure	Schema compatibility verified via SQLAlchemy; zero API changes	**APPROVED**	Sprint 4

				changes		

Phase 19: Change Request Log & Change Governance

CR ID	Change Title	Requester	Classification	Impact Summary	Decision Status	Decision Date
				required.		
CR-002	Adjust Free Deduplication Tier from 5 GB to 10 GB	Product Owner	Business / Monetization	Increased database initial credit column; lowers onboarding friction.	**APPROVED**	Sprint 4
CR-003	Cloud API Direct In-Place Crawling (Google Drive / OneDrive)	Enterprise Sales	Product Feature Scope	**[SIMULATED INTERVIEW SCENARIO]** High complexity; deferred to Phase 3 Roadmap.	**DEFERRED**	Sprint 5 Planning
CR-004	Hard-Link Deduplication Mode on NTFS / APFS / EXT4	IT Admin User	Core Engine Feature	**[SIMULATED INTERVIEW SCENARIO]** Replaces duplicates with OS hard links instead of archiving.	**APPROVED (For v1.1)**	Sprint 5 Planning

3. Detailed Change Requests

CR-002: Adjust Free Deduplication Credit from 5 GB to 10 GB [ACTUAL PROJECT EVENT]

- **Requester:** Product Owner (Marcus)
- **Description:** Increase the default credit byte grant on user registration from 5 GB ($5,368,709,120$ bytes) to 10 GB ($10,737,418,240$ bytes).
- **Business Rationale:** Strategy documents originally proposed 5 GB, but early prosumer user feedback indicated that photographers shoot average batches of 8-10 GB per session. A 5 GB limit blocked users mid-scan during their very first evaluation, causing premature churn.
- **BA Impact Assessment:**
- *Requirement Impact:* Updates BR-003, FR-012, and schema in `backend/app/models.py#L13`.
- *Engineering Effort:* Minimal (1 line modification in SQLAlchemy model default parameter).
- *Revenue Impact:* Increases customer trust and allows prospective users to experience a complete cleanup workflow before purchasing a $4.99 credit pack.
- **CCB Decision: APPROVED unanimously**.

CR-003: Cloud API Direct In-Place Crawling [SIMULATED INTERVIEW SCENARIO]

- **Requester:** Enterprise Sales Representative
- **Description:** Implement direct cloud API connectors allowing Duplicate Bin SaaS to scan Google Drive, Dropbox, and OneDrive accounts without requiring files to be synced to a local computer drive.
- **Business Rationale:** Multiple prospective enterprise clients want to clean up cloud storage directly from their web browser.
- **BA Impact Assessment:**
- *Requirement Impact:* Adds new module `CloudConnectorEngine`, requires OAuth2 multi-tenant cloud credential storage, background cloud worker clusters, and handles cloud API rate limiting (Google Drive API quota).
- *Engineering Effort:* Estimated 180 story points (6 sprints); requires dedicated cloud infrastructure budget.
- *Scope Evaluation:* Deviates significantly from current v0.2.0 local desktop architecture.

- **CCB Decision: DEFERRED to Phase 3 Enterprise Roadmap**. Current users are advised to mount cloud drives as local virtual file streams (Google Drive for Desktop / OneDrive Files On-Demand).

CR-004: Hard-Link Deduplication Mode [SIMULATED INTERVIEW SCENARIO]

- **Requester:** Corporate IT Administrator (Marcus)
- **Description:** Add an alternative cleanup action to "Archive" called "Create Hard Links". When selected, redundant file copies are replaced with operating system hard links (`os.link`) pointing to the master file's physical disk sectors.
- **Business Rationale:** On shared office servers, existing project directory references remain valid and accessible, but disk consumption drops to the size of a single copy.
- **BA Impact Assessment:**
- *Requirement Impact:* Adds new selection action in `archive.py` and UI dialog.
- *Technical Constraint:* Hard links only function across the same physical disk partition; cannot cross filesystems.
- *Engineering Effort:* Moderate (20 story points; 1 sprint).
- **CCB Decision: APPROVED for release in Milestone v1.1.0**.